Part 2: Quantitative data analysis

Quantitative Session of the 2026 HUSOE ELPS Methods and Data Institute

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0.0.1 The Big Picture

Prior to jumping into more details about univariate data examples in the next section, it will be important to gather a "big picture" view of the field. This big picture view will provide us with a high-level description of the various topics we will cover in the course.

0.0.1.1 Population parameter

A number which summarizes the entire group.

0.0.1.2 Sample statistic

A single number that summarizes a subset of data, or the sample.

0.0.1.3 Confidence interval (CI)

- Range of likely or plausible values for a population parameter.
- Based on a sample and statistics from that sample.
- *Margin of error* represents the number of standard deviations on a statistic.

0.0.1.4 Hypothesis testing

Statistical procedure used to test an existing claim about a population.

- Test is based on data; most ideally data collected via probability sampling.
- H_0 - Null hypothesis: If data supports the claim, fail to reject H_0
- H_a - Alternative hypothesis: If data does not support claim, reject H_0

0.0.1.5 Analysis of variance (ANOVA)

Comparing means of more than two populations.

F-statistic is a ratio that is used to compare variability between sets.

0.0.1.5.1 Multiple comparison procedures

Set of statistical tests that compare means to each other.

- Examples include Turkey's test, Least significant difference (LSD), pairwise t-test
- These tests are only conducted if you analysis of variance identifies differences

0.0.1.6 Interaction effects

Interaction effects are relevant to statistical models that use two or more variables.

0.0.1.7 Correlation

Measures the strength and direction of a linear relationship between two variables.

0.0.1.8 Linear regression

Helps make predictions for one variable based on the values of another.

- There are many types of regression:
  - Simple linear regression
  - Multiple linear regression
  - Logistic regression
  - Non-linear regression

0.0.1.9 Chi-square test

When using correlation and regression analyses, one core assumption is that the variables are quantitative in nature.

We use a chi-square test to study categorical variables.

0.1 Case Study: HBCUs and Educational Geography

In this section, we analyze data on Historically Black Colleges and Universities (HBCUs) to explore patterns in location, institutional type, and founding history.

0.1.1 Step 1: Load Packages

```
#install.packages("tidyverse")
#install.packages("lubridate")
library(tidyverse)
library(lubridate)
```

We'll start with a guided example using data on Historically Black Colleges and Universities.

0.1.2 Step 2: Load the data

```
library(tidyverse)
hbcu <- read_csv("https://raw.githubusercontent.com/quant-shop/intro-comp-educ-soc/refs/heads/m

# View the first few rows
head(hbcu)
```

```
# A tibble: 6 x 7
  name                       city       state founded   lat   lon type
  <chr>                      <chr>      <chr>   <dbl> <dbl> <dbl> <chr>
1 Alabama A&M University     Normal     AL       1875  34.8 -86.6 Public, 4 Year
2 Alabama State University   Montgomery AL       1867  32.4 -86.3 Public, 4 Year
3 Albany State University    Albany     GA       1903  31.6 -84.2 Public, 4 Year
4 Alcorn State University    Lorman     MS       1871  31.9 -91.1 Public, 4 Year
5 Allen University           Columbia   SC       1870  34.0 -81.0 Private, 4 Year
6 American Baptist College   Nashville  TN       1924  36.2 -86.8 Private, 4 Year
```

```r
# Check structure
glimpse(hbcu)
```

```
Rows: 102
Columns: 7
$ name    <chr> "Alabama A&M University", "Alabama State University", "Albany ~
$ city    <chr> "Normal", "Montgomery", "Albany", "Lorman", "Columbia", "Nashv~
$ state   <chr> "AL", "AL", "GA", "MS", "SC", "TN", "AR", "SC", "NC", "FL", "A~
$ founded <dbl> 1875, 1867, 1903, 1871, 1870, 1924, 1884, 1870, 1873, 1904, 19~
$ lat     <dbl> 34.7834, 32.3643, 31.5785, 31.8769, 34.0298, 36.1659, 34.7465,~
$ lon     <dbl> -86.5683, -86.2952, -84.1543, -91.1458, -81.0115, -86.7844, -9~
$ type    <chr> "Public, 4 Year", "Public, 4 Year", "Public, 4 Year", "Public,~
```

```r
# Summary statistics
summary(hbcu)
```

```
     name               city              state              founded    
 Length:102         Length:102         Length:102         Min.   :1837  
 Class :character   Class :character   Class :character   1st Qu.:1870  
 Mode  :character   Mode  :character   Mode  :character   Median :1886  
                                                          Mean   :1895  
                                                          3rd Qu.:1905  
                                                          Max.   :1988  
      lat             lon              type          
 Min.   :18.34   Min.   :-98.50   Length:102        
 1st Qu.:32.48   1st Qu.:-90.13   Class :character  
 Median :34.02   Median :-84.64   Mode  :character  
 Mean   :34.31   Mean   :-85.13                     
 3rd Qu.:36.17   3rd Qu.:-80.78                     
 Max.   :39.93   Max.   :-64.96                     
```

0.1.3 Step 3: Clean the data

```r
hbcu <- hbcu %>%
  mutate(
    founded = as.numeric(founded),
    type = as.factor(type),
    state = as.factor(state)
  )
```

0.1.4 Step 4: Basic Exploration

How many institutions are in the data set?

```r
nrow(hbcu)
```

```
[1] 102
```

What is the distribution by HBCU type?

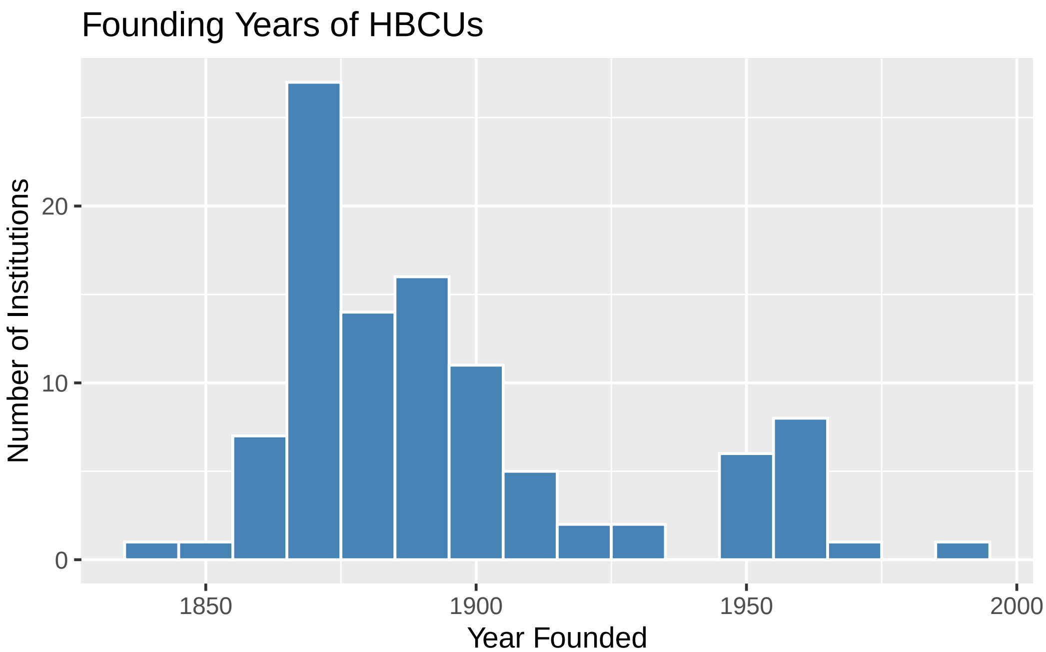
```r
hbcu %>%
  count(type) %>%
  arrange(desc(n))
```

```
# A tibble: 5 x 2
  type                    n
  <fct>               <int>
1 Private, 4 Year        45
2 Public, 4 Year         40
3 Public, 2 Year         11
4 Private, Specialized    4
5 Private, 2 Year         2
```

0.1.5 Step 5: Historical Analysis

When were the HBCUs founded?

```r
hbcu %>%
  ggplot(aes(x = founded)) +
  geom_histogram(binwidth = 10, fill = "steelblue", color = "white") +
  labs(
    title = "Founding Years of HBCUs",
    x = "Year Founded",
    y = "Number of Institutions"
  )
```



0.1.6 Step 6: Geographic Distribution

HBCUs by state

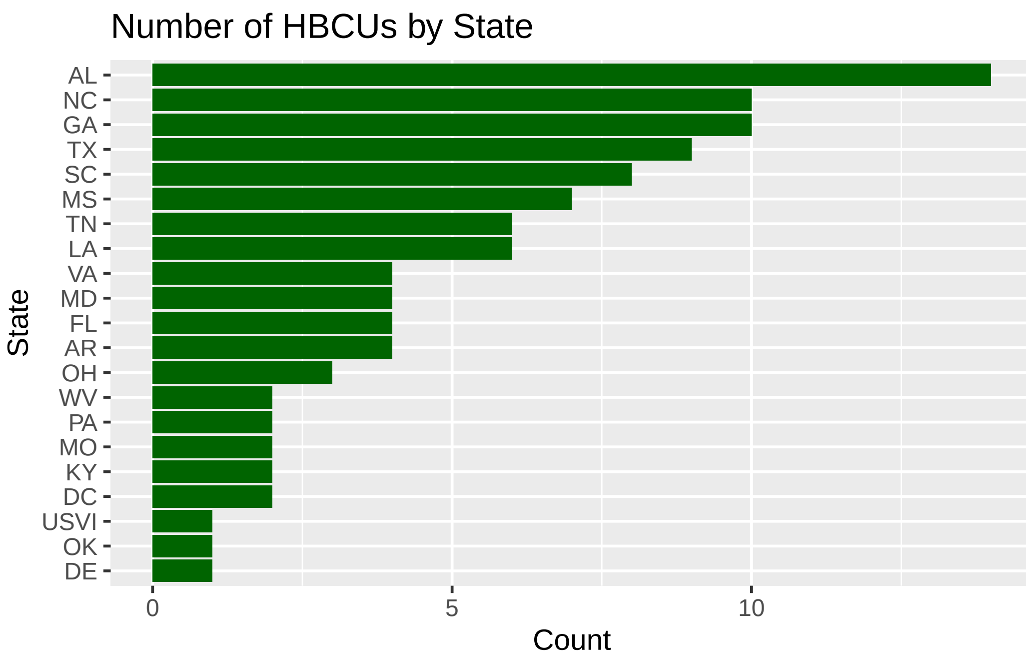
```
hbcu %>%
  count(state, sort = TRUE)
```

```
# A tibble: 21 x 2
   state     n
   <fct> <int>
 1 AL       14
 2 GA       10
 3 NC       10
 4 TX        9
 5 SC        8
 6 MS        7
 7 LA        6
 8 TN        6
 9 AR        4
10 FL        4
# i 11 more rows
```

Basic visualization of HBCUs by state

```
hbcu %>%
  count(state) %>%
  ggplot(aes(x = reorder(state, n), y = n)) +
```

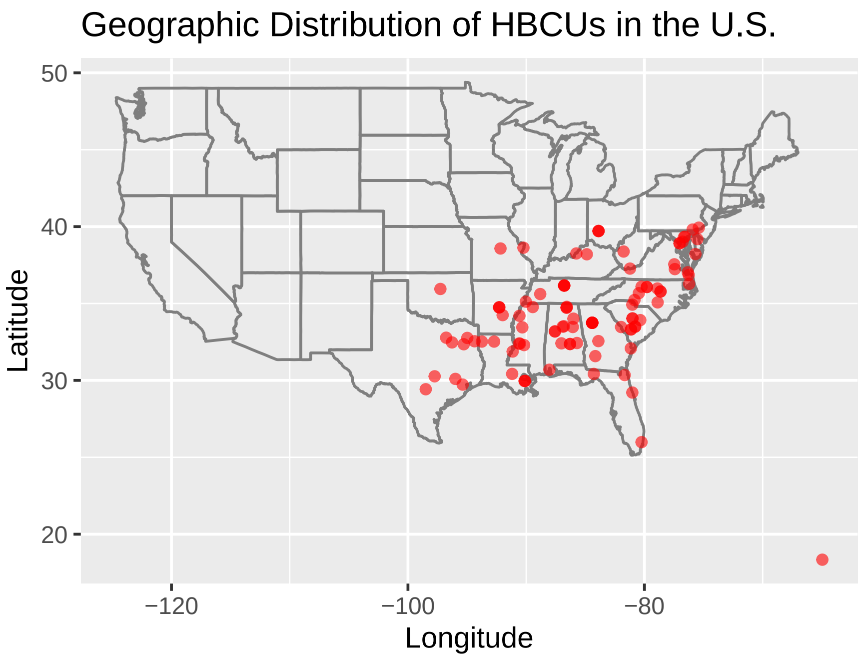
```r
  geom_col(fill = "darkgreen") +
  coord_flip() +
  labs(
    title = "Number of HBCUs by State",
    x = "State",
    y = "Count"
  )
```



0.1.7 Step 7: Mapping

```r
#install.packages("maps", repos = "http://cran.us.r-project.org")
library(maps)
```

```r
hbcu %>%
  ggplot(aes(x = lon, y = lat)) +
  borders("state") +
  geom_point(color = "red", alpha = 0.6) +
  coord_fixed(1.3) +
  labs(
    title = "Geographic Distribution of HBCUs in the U.S.",
    x = "Longitude",
    y = "Latitude"
  )
```



0.2 Case Study: Data on State School Districts

0.2.1 Step 1: Load the libraries

```
library(dplyr)
library(readr)
library(stringr)
library(janitor)
```

0.2.2 Step 2: Load the data

```
district <- read.csv("https://raw.githubusercontent.com/quant-shop/intro-comp-educ-soc/refs/hea
# district <- read.csv("../data/state-district-data.csv")

str(district)
```

```
'data.frame':	57 obs. of  36 variables:
 $ State.Name                                                   : chr  "ALABAMA" "ALASKA"
 $ State.Abbr                                                   : chr  "AL " "AK " "AS "
 $ ANSI.FIPS.State.Code..State..Latest.available.year           : int  1 2 60 4 5 59 6 8
 $ Total.Number.Operational.School.Districts..District..2023.24 : int  156 54 1 685 304 
 $ Total.Number.Operational.Schools..Public.School..2023.24     : int  1524 494 29 2427 
```

```
 $ Total.Number.Operational.Charter.Schools..Public.School..2023.24   : chr  "18" "31" "†" "57
 $ Prekindergarten.Students..State..2023.24                           : chr  "21924" "3320" "-
" "20810" ...
 $ Kindergarten.Students..State..2023.24                              : chr  "55881" "9257" "-
" "72320" ...
 $ Grades.1.8.Students..State..2023.24                                : chr  "447317" "78890" 
" "654843" ...
 $ Grades.9.12.Students..State..2023.24                               : chr  "223528" "39776" 
" "369124" ...
 $ Grade.13.Students..State..2023.24                                  : chr  "†" "†" "†" "†" .
 $ Ungraded.Students..State..2023.24                                  : chr  "†" "†" "†" "533"
 $ American.Indian.Alaska.Native.Students..State..2023.24             : chr  "5997" "28284" "-
" "46878" ...
 $ Asian.or.Asian.Pacific.Islander.Students..State..2023.24           : chr  "11016" "6458" "-
" "35975" ...
 $ Hispanic.Students..State..2023.24                                  : chr  "84196" "10076" "
" "539087" ...
 $ Black.or.African.American.Students..State..2023.24                 : chr  "236304" "2955" "
" "64938" ...
 $ White.Students..State..2023.24                                     : chr  "381550" "61502" 
" "378320" ...
 $ Nat..Hawaiian.or.Other.Pacific.Isl..Students..State..2023.24       : chr  "831" "4148" "-
" "4017" ...
 $ Two.or.More.Races.Students..State..2023.24                         : chr  "28756" "17820" "
" "48415" ...
 $ Total.Enrollment..Exclude.AE..for.SY.2014.15.onward..State..2023.24: chr  "748650" "131243"
" "1117630" ...
 $ Full.Time.Equivalent..FTE..Teachers..State..2023.24                : chr  "42857.92" "7221.
" "49260.47" ...
 $ Pupil.Teacher.Ratio..State..2023.24                                : chr  "17.47" "18.17" "
" "22.69" ...
 $ State.Name..State..2023.24                                         : chr  "ALABAMA" "ALASKA
 $ Total.Number.of.Districts.with.Enrollment..District..2023.24       : int  150 54 0 664 260 
 $ Total.Number.of.School.Districts..District..2023.24                : int  159 54 1 722 308 
 $ Total.Number.of.Public.Schools..Public.School..2023.24             : int  1527 498 29 2557 
 $ Male.Students..State..2023.24                                      : chr  "384500" "67646" 
 $ Female.Students..State..2023.24                                    : chr  "364150" "63597" 
 $ Black.or.African.American...male..State..2023.24                   : chr  "120605" "1488" "
" "33191" ...
 $ Black.or.African.American...female..State..2023.24                 : chr  "115699" "1467" "
" "31747" ...
 $ White...male..State..2023.24                                       : chr  "197384" "31786" 
" "194292" ...
 $ White...female..State..2023.24                                     : chr  "184166" "29716" 
" "184028" ...
 $ Grade.12.Students...Black.or.African.American..State..2023.24      : chr  "15413" "241" "-
" "5876" ...
 $ Secondary.Teachers..State..2023.24                                 : chr  "20188.71" "3673.
```

```
" "15996.72" ...
 $ Elementary.Teachers..State..2023.24                                   : chr  "18395.46" "2985.0
" "30478.09" ...
 $ Full.Time.Equivalent..FTE..Staff..State..2023.24                      : chr  "79963.90" "16423
" "109199.43" ...
```

0.2.3 Step 3: Clean the data

```r
district_clean <- district %>%

  # Fix missing/suppressed values
  mutate(across(where(is.character), ~na_if(., "-"))) %>%
  mutate(across(where(is.character), ~na_if(., "†"))) %>%
  mutate(across(where(is.character), str_trim)) %>%

  # Convert numeric columns safely
  mutate(across(
    where(is.character),
    ~parse_number(.)
  ))
```

0.2.4 Step 4: Feature engineering

We can create variables that answer educational questions.

```r
str(district_clean)
```

```
'data.frame':   57 obs. of  36 variables:
 $ State.Name                                                            : num  NA NA NA NA NA NA
  ..- attr(*, "problems")= tibble [57 x 4] (S3: tbl_df/tbl/data.frame)
  .. ..$ row     : int [1:57] 1 2 3 4 5 6 7 8 9 10 ...
  .. ..$ col     : int [1:57] NA NA NA NA NA NA NA NA NA NA ...
  .. ..$ expected: chr [1:57] "a number" "a number" "a number" "a number" ...
  .. ..$ actual  : chr [1:57] "ALABAMA" "ALASKA" "AMERICAN SAMOA" "ARIZONA" ...
 $ State.Abbr                                                            : num  NA NA NA NA NA NA
  ..- attr(*, "problems")= tibble [57 x 4] (S3: tbl_df/tbl/data.frame)
  .. ..$ row     : int [1:57] 1 2 3 4 5 6 7 8 9 10 ...
  .. ..$ col     : int [1:57] NA NA NA NA NA NA NA NA NA NA ...
  .. ..$ expected: chr [1:57] "a number" "a number" "a number" "a number" ...
  .. ..$ actual  : chr [1:57] "AL" "AK" "AS" "AZ" ...
 $ ANSI.FIPS.State.Code..State..Latest.available.year                    : int  1 2 60 4 5 59 6 8
 $ Total.Number.Operational.School.Districts..District..2023.24          : int  156 54 1 685 304 1
 $ Total.Number.Operational.Schools..Public.School..2023.24              : int  1524 494 29 2427 1
 $ Total.Number.Operational.Charter.Schools..Public.School..2023.24      : num  18 31 NA 571 105 1
 $ Prekindergarten.Students..State..2023.24                              : num  21924 3320 NA 2081
```

```
 $ Kindergarten.Students..State..2023.24                                : num  55881 9257 NA 7232
 $ Grades.1.8.Students..State..2023.24                                  : num  447317 78890 NA 65
 $ Grades.9.12.Students..State..2023.24                                 : num  223528 39776 NA 36
 $ Grade.13.Students..State..2023.24                                    : num  NA NA NA NA NA NA
 $ Ungraded.Students..State..2023.24                                    : num  NA NA NA 533 174 N
 $ American.Indian.Alaska.Native.Students..State..2023.24               : num  5997 28284 NA 4687
 $ Asian.or.Asian.Pacific.Islander.Students..State..2023.24             : num  11016 6458 NA 3597
 $ Hispanic.Students..State..2023.24                                    : num  84196 10076 NA 539
 $ Black.or.African.American.Students..State..2023.24                   : num  236304 2955 NA 649
 $ White.Students..State..2023.24                                       : num  381550 61502 NA 37
 $ Nat..Hawaiian.or.Other.Pacific.Isl..Students..State..2023.24         : num  831 4148 NA 4017 4
 $ Two.or.More.Races.Students..State..2023.24                           : num  28756 17820 NA 484
 $ Total.Enrollment..Exclude.AE..for.SY.2014.15.onward..State..2023.24: num  748650 131243 NA 1
 $ Full.Time.Equivalent..FTE..Teachers..State..2023.24                  : num  42858 7221 NA 4926
 $ Pupil.Teacher.Ratio..State..2023.24                                  : num  17.5 18.2 NA 22.7
 $ State.Name..State..2023.24                                           : num  NA NA NA NA NA NA
  ..- attr(*, "problems")= tibble [57 x 4] (S3: tbl_df/tbl/data.frame)
  .. ..$ row     : int [1:57] 1 2 3 4 5 6 7 8 9 10 ...
  .. ..$ col     : int [1:57] NA NA NA NA NA NA NA NA NA NA ...
  .. ..$ expected: chr [1:57] "a number" "a number" "a number" "a number" ...
  .. ..$ actual  : chr [1:57] "ALABAMA" "ALASKA" "AMERICAN SAMOA" "ARIZONA" ...
 $ Total.Number.of.Districts.with.Enrollment..District..2023.24         : int  150 54 0 664 260 1
 $ Total.Number.of.School.Districts..District..2023.24                  : int  159 54 1 722 308 1
 $ Total.Number.of.Public.Schools..Public.School..2023.24               : int  1527 498 29 2557 1
 $ Male.Students..State..2023.24                                        : num  384500 67646 NA 56
 $ Female.Students..State..2023.24                                      : num  364150 63597 NA 54
 $ Black.or.African.American...male..State..2023.24                     : num  120605 1488 NA 331
 $ Black.or.African.American...female..State..2023.24                   : num  115699 1467 NA 317
 $ White...male..State..2023.24                                         : num  197384 31786 NA 19
 $ White...female..State..2023.24                                       : num  184166 29716 NA 18
 $ Grade.12.Students...Black.or.African.American..State..2023.24        : num  15413 241 NA 5876
 $ Secondary.Teachers..State..2023.24                                   : num  20189 3673 NA 1599
 $ Elementary.Teachers..State..2023.24                                  : num  18395 2985 NA 3047
 $ Full.Time.Equivalent..FTE..Staff..State..2023.24                     : num  79964 16423 NA 109
  ..- attr(*, "problems")= tibble [1 x 4] (S3: tbl_df/tbl/data.frame)
  .. ..$ row     : int 51
  .. ..$ col     : int NA
  .. ..$ expected: chr "a number"
  .. ..$ actual  : chr "‡"
```

```
# what kinds of questions might you answer with this data set?
```

0.3 Recommended readings

Knaflic, C. N. (2015). *Storytelling with data: A data visualization guide for business professionals.* Wiley.